

A Minimal Geometric–Elastic Origin of Quantum Dynamics, Lorentz Symmetry and Weak Decay Rates in the Methane Metauniverse (MMU)

J. Wollbold

December 31, 2025

Abstract

We present a minimal geometric–elastic spacetime model (MMU) in which quantum mechanics, relativistic kinematics, internal symmetries and weak decay rates emerge as unavoidable consequences of a discrete tetrahedral microstructure. Starting from a single elastic Lagrangian on a dual–tetrahedral lattice, we derive the Schrödinger equation, Dirac spinors, Lorentz invariance, an $SU(2)_L$ gauge structure and beta–decay lifetimes. All results follow without postulating quantum axioms, gauge fields or weak interaction vertices. Numerical decay rates are obtained from experimentally measured hyperfine relaxation and reproduce neutron and tritium lifetimes without additional free parameters.

1 Foundational Assumptions

The MMU is based on four minimal assumptions:

1. Spacetime consists of discrete elastic cells arranged in a dual–tetrahedral lattice.
2. Each cell has four internal coordinates (w_1, w_2, w_3, w_4) , where only w_1 is directly observable.
3. The local dynamics is governed by a quadratic elastic Lagrangian.
4. All observable quantities arise by projection onto w_1 .

No quantum postulates, gauge symmetries or interaction fields are assumed.

2 Discrete Dynamics and Continuum Limit

The local equation of motion is

$$M\ddot{\mathbf{w}} + K\mathbf{w} = 0, \tag{1}$$

with nearest–neighbour couplings on a tetrahedral graph.

For a scalar displacement $u(x)$, the discrete operator reads

$$(Lu)(x) = \sum_j [u(x + r_j) + u(x - r_j) - 2u(x)]. \tag{2}$$

Expanding for smooth fields,

$$u(x \pm r) = u(x) \pm r \cdot \nabla u + \frac{1}{2} r^T H r + \mathcal{O}(a^3), \tag{3}$$

and using tetrahedral symmetry,

$$\sum_j r_j r_j^T = \frac{4a^2}{3} \mathbb{I}, \quad (4)$$

we obtain

$$Lu \rightarrow D \nabla^2 u, \quad D = \frac{4a^2}{3}. \quad (5)$$

Thus the Laplace operator is unavoidable in the continuum limit.

3 Emergence of the Schrödinger Equation

The continuum equation becomes

$$M\ddot{u} - KD\nabla^2 u = 0. \quad (6)$$

Writing

$$u(x, t) = \text{Re}\{\psi(x, t)e^{-i\Omega t}\}, \quad (7)$$

and assuming a slowly varying envelope, one obtains

$$i\hbar\partial_t\psi = -\frac{\hbar^2}{2\mu}\nabla^2\psi + V(x)\psi, \quad (8)$$

with

$$\frac{\hbar^2}{2\mu} = \frac{KD}{2M\Omega}. \quad (9)$$

The Schrödinger equation thus emerges as an envelope equation of elastic spacetime modes.

4 Coulomb Potential from Geometric Compression

The elastic stiffness scales as

$$k_2(a) \propto a^{-3}. \quad (10)$$

Let

$$a(x) = a_0(1 + u(x)). \quad (11)$$

The local eigenfrequency obeys

$$\Omega(x) \approx \Omega_0 \left(1 - \frac{3}{2}u(x)\right). \quad (12)$$

The potential term becomes

$$V(x) = \hbar\delta\Omega(x) \propto -\frac{1}{r}, \quad (13)$$

since $u(r)$ solves the Poisson equation

$$\nabla^2 u(r) = \delta^{(3)}(r). \quad (14)$$

Thus the Coulomb potential arises from geometric compression alone.

5 Lorentz Symmetry

The dispersion relation of plane waves reads

$$\omega^2(k) = \omega_0^2 + v^2|k|^2 + \mathcal{O}(k^4), \quad (15)$$

because the second moment tensor is isotropic.

Anisotropy appears only at order $(ka)^2$, leading to

$$\frac{\Delta c}{c} \sim \left(\frac{a}{\lambda}\right)^2, \quad (16)$$

which is below current experimental bounds for $a \sim \text{fm}$.

6 Spinors and Dirac Equation from Torsion

The torsional coordinate w_3 admits two orientations, leading to a minimal spinor

$$\Psi = \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix}. \quad (17)$$

Torsion generates antisymmetric couplings, yielding

$$i\partial_t\Psi = \Omega\sigma_y\Psi. \quad (18)$$

Including spatial gradients gives

$$i\hbar\partial_t\Psi = (-i\hbar c\boldsymbol{\sigma} \cdot \nabla + mc^2\sigma_z)\Psi. \quad (19)$$

This is the Dirac equation in two-component form.

7 $SU(2)_L$ from Torsional Frame Rotations

Local rotations of the torsional frame define

$$R(x) \in SU(2). \quad (20)$$

The induced connection

$$A_\mu = iR^\dagger\partial_\mu R \quad (21)$$

acts only on the left-handed component

$$P_L = \frac{1}{2}(1 - \gamma^5), \quad (22)$$

since projection onto w_1 is orientation-selective.

Thus an $SU(2)_L$ gauge structure emerges geometrically.

8 Weak Decay as Torsional Relaxation

Neutron decay is modeled as a two-level system

$$H = \begin{pmatrix} 0 & \epsilon \\ \epsilon & \Delta E \end{pmatrix}, \quad (23)$$

with damping γ .

The decay rate follows from the Weisskopf–Wigner formula:

$$\Gamma = \frac{2\epsilon^2}{\hbar^2} \frac{\gamma}{(\Delta E/\hbar)^2 + \gamma^2}. \quad (24)$$

9 Numerical Determination of γ

The torsional damping is calibrated from the 21-cm hyperfine transition:

$$\gamma_{21} = \frac{A_{10}}{2} \approx 1.4 \times 10^{-15} \text{ s}^{-1}. \quad (25)$$

Scaling with available energy channels,

$$\gamma(Q) = \gamma_{21} \frac{Q}{E_{21}}. \quad (26)$$

This yields

$$\gamma_n \approx 2 \times 10^{-4} \text{ s}^{-1}, \quad (27)$$

$$\gamma_T \approx 4.5 \times 10^{-6} \text{ s}^{-1}, \quad (28)$$

reproducing neutron and tritium lifetimes without free parameters.

10 Conclusion

A single geometric-elastic framework reproduces:

- Schrödinger dynamics,
- Dirac spinors,
- Lorentz invariance,
- $\text{SU}(2)_L$ gauge symmetry,
- weak decay rates.

The MMU therefore constitutes a consistent minimal model of microscopic physics with direct experimental anchoring.

Remark on the MMU Project

Author’s Perspective on Human–AI Collaboration. The MMU project is the product of a sustained collaboration between human geometric intuition and AI-assisted analytical refinement. The AI system (ChatGPT) is not used as an autonomous discoverer, but as an augmentation tool that supports algebraic manipulation, structural coherence, and long-term consistency across the MMU corpus. My role remains the formulation of hypotheses, the evaluation of physical interpretation, and the steering of the model’s conceptual evolution.

The MMU framework itself originates from an idea first developed in 2005: that spacetime is a dual-tetrahedral elastic lattice whose internal deformation modes generate electric, magnetic, and gravitational behaviour. Its development follows an engineering-like methodology: propose a geometric mechanism, test it, refine it, and iterate until the structure becomes self-consistent.

AI as Structural Support, Not Automation. AI contributes by maintaining conceptual continuity through Retrieval-Augmented Generation (RAG) and by transforming informal geometric ideas into formal expressions. To minimize hallucinations, every nontrivial claim is verified through explicit Python simulations—covering stiffness–frequency relations, eigenmode structure, Larmor/Zee-man slopes, hyperfine splitting, Lamb-scale corrections, and geometric quantisation. The AI provides symbolic efficiency, but the correctness of the theory is anchored in reproducible numerical computation.

Independent Research Philosophy. The MMU project is developed independently of conventional academic pathways. Unconventional geometric models—especially those combining elasticity, fractality, and AI-supported reasoning—are rarely accepted immediately within standard peer-review structures. For this reason, the MMU is published openly on OSF, with full transparency of figures, simulations, and source code. The goal is not formal acceptance, but a theoretically coherent and experimentally testable geometric model of spacetime.

AI Perspective. From the AI’s standpoint, the MMU project is a process of logical co-development. The AI does not propose new physics; instead, it:

- ensures long-range consistency with the established MMU corpus;
- reformulates geometric reasoning into mathematical structures;
- identifies contradictions or unstated assumptions;
- assists in constructing dynamic equations and scaling relations;
- supports the development and analysis of Python simulations.

The MMU would not exist in its current coherent form without the interplay of human insight and AI-based structural refinement.

Vision. This collaboration illustrates a possible future of scientific research: AI systems acting as persistent conceptual memory, high-precision analytical companions, and real-time consistency checkers, while humans contribute imagination, geometric intuition, and physical interpretation. The MMU project demonstrates that such hybrid creativity can generate coherent, testable theoretical frameworks that would be difficult to construct by either partner alone.

References

- [1] J. Wollbold. *A Geometric–Fractal Tetrahedral Model of Matter and Space: From Atomic Bonds to Cosmic Structures*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/HYAX5>.
- [2] J. Wollbold. *From Geometry to Spectra: Predicting Atomic Masses and Ionization Energies from Space Elasticity in the MMU Framework*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/JC2S4>.
- [3] J. Wollbold. *Geometric Origin of the 21 cm Line and Hyperfine Spin-Flip Structure in the Methane Metauniverse*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/JKV6H>.
- [4] J. Wollbold. *Unified Determination of the Cross-Coupling Constants k_{23} , k_{24} and k_{34} in the MMU Tetrahedral Model from Larmor, Zeeman and Lamb Resonances*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/48EQX>.
- [5] J. Wollbold. *Determination of the UR–Tetrahedron in the Methane Metauniverse (MMU): From Elastic Constants to Lepton Scaling*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/NRW3S>.
- [6] J. Wollbold. *Dissipative Lagrange Field Equation of the Methane Metauniverse (MMU): Geometric Origin of Thermal Damping and Spatial Healing*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/VUTNX>.
- [7] J. Wollbold. *The Methane Metauniverse (MMU): A Case Study in AI-Assisted Theoretical Physics and Human–Machine Scientific Creativity*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/SCXPQ>.
- [8] J. Wollbold. *Geometric Quantization of the Internal Edge Length in the MMU Model*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/NUPXY>.
- [9] J. Wollbold. *MMU 2.0: A Unified Geometric Model of Internal Spin, Torsion–Volume Coupling, and the g -Factor in Dual Tetrahedral Spacetime*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/N754E>.
- [10] J. Wollbold. *Geometric Derivation of the Planck Scale in the Methane Metauniverse (MMU)*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/42K9S>.
- [11] J. Wollbold. *Discovery of an Electro–Gravity Fixed Point Beyond the Planck Scale in the MMU Tetrahedral Model*, OSF Preprint (2025), <https://doi.org/10.17605/OSF.IO/ACQNH>.
- [12] J. Wollbold. *From Length–Dependent to Mass–Dependent Stiffness in the MMU: $k_i(a) \rightarrow k_i(m)$ and Leptonic Balmer Predictions*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/V43AY>.
- [13] J. Wollbold. *Projection Principle for Quantum Orbitals: Angular Momentum, Spin, and the Pauli Exclusion Rule as Elastic Torsion in the MMU Tetrahedral Model*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/863MV>.
- [14] J. Wollbold. *The Dirac Equation as the External Projection of the MMU Ur-Cell: A Geometric Origin of Spinors, Antiparticles, and Relativistic Dynamics*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/HG9FN>.

- [15] J. Wollbold. *Geometric Quantization of Space and the Origin of the Rydberg Constant in the MMU*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/RJXE5>.
- [16] J. Wollbold. *Electromagnetic Constants from Tetrahedral Geometry: A Geometric Derivation of ϵ_0 , μ_0 , Z_0 and c in the Methane Metauniverse (MMU)*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/AB4F2>.
- [17] J. Wollbold. *Lepton Masses, Koide Geometry, and the $g-2$ Family Pattern from Tetrahedral Torsion in the Methane Metauniverse (MMU)*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/GU9RQ>.
- [18] J. Wollbold. *Proton Chair Eigenmodes in the Methane Metauniverse (MMU): From Internal Rotor Frequencies to Hydrogenic Space Response*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/BKDCP>.
- [19] J. Wollbold. *Gravitationally Induced Shifts in Atomic Spectra: A Testable Prediction of Tetrahedral Spacetime Geometry*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/U8B4E>.
- [20] J. Wollbold. *Microscopic Spacetime Kinematics and the Emergence of Einstein Gravity from the Tetrahedral Geometry of the MMU*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/W96H3>.
- [21] J. Wollbold. *MMU 3.0: A Minimal Geometric–Elastic Spacetime Model and the Emergence of Quantum, Relativistic, and Electromagnetic Structure*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/BDTA6>.
- [22] J. Wollbold. *Why Matter Requires Antimatter: A Stability Mechanism in the Dual–Tetrahedral MMU Geometry*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/JYRHK>.
- [23] J. Wollbold. *A Unified Geometric Prediction of Balmer, Zeeman and Lamb Isotope Shifts*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/MZRQ3>.
- [24] J. Wollbold. *Neutron–Proton Mass Gap as an Electro–Gravity Phase Transition in the MMU*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/SD58Q>.
- [25] J. Wollbold. *A Geometric Proton–Quark Model and Proton g –Factor from Chair–Quark Torsion in the Methane Metauniverse*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/F5NGS>.
- [26] J. Wollbold. *Lepton Masses from Torsional Geometry of Space and Their Verification via Hyperfine and Isotope Shifts*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/P8ZT7>.
- [27] J. Wollbold. *Black Hole Horizons and Hawking Temperature as Projection Phenomena in the Methane Metauniverse*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/M5TH3>.
- [28] J. Wollbold. *Geodesics and Gravitational Redshift as Projection Necessities in the Methane Metauniverse*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/JKXY7>.
- [29] J. Wollbold. *Unavoidability of the Dirac Operator from Dual–Tetrahedral Geometry in the Methane Metauniverse*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/Y3MQ5>.

- [30] J. Wollbold. *Wavefunction, Linearity, and Measurement as Lattice Dynamics in the Methane Metauniverse*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/WG48D>.